Latency Can Reverse the Ranking: Effective-Rate Evaluation of RIS Phase Optimization for Mobile Users

Seda Savaşçı Şen, Ali Çalhan, and Murtaza Cicioğlu

***Abstract*—Reconfigurable intelligent surface (RIS) phase optimizers are usually compared on the channel they were given, but a mobile user receives a phase only after the optimizer has finished. We evaluate methods by this measured-latency *effective rate* in a 3GPP indoor-office multi-user MISO downlink with a 32-element RIS and users walking at 1.5 m/s at 3 GHz. Coordinate descent and long-run Adam, the best methods at $t = 0$ at their default budgets, need 0.9–5.9 s and end below a 2.5 ms single-user heuristic. Adam's effective rate peaks at 4–6 steps, and running it to 48 steps costs 14.0% (95% CI 12.5–15.5%). The optimum is governed by the step time relative to the Doppler frequency, x: a small-delay model explains its scaling, $x^{-2/(\gamma+2)}$, and the measured exponent (-0.675) lies in the model's range; at the standardized indoor speeds of 3 and 10 km/h the optimum is 8 and 3 steps. With periodic re-optimization, the pilot overhead sets the update period, and updates of at most 8 steps suffice. A learned update rule and a published phase-aware CNN–LSTM, ported to a single user, are both dominated by faster Adam configurations. All results assume a precoder recomputed from the current channel.**

***Index Terms*—Reconfigurable intelligent surface, mobility, computation latency, effective rate, channel aging, deep unfolding, multi-user MISO.**

I. Introduction

RECONFIGURABLE intelligent surfaces (RIS) reshape the radio channel with a large number of passive, phase-controllable elements [1]–[4], and prototypes and field trials have demonstrated them in practice [5], [6]. In a multi-user downlink the phases must be optimized jointly with the base-station (BS) precoder, which is a non-convex problem usually attacked with alternating optimization and semidefinite relaxation, block coordinate descent, fractional programming or WMMSE [2], [7]–[10], and, more recently, with learned policies [11].

These methods are almost always compared by the rate they reach on the channel they were given. That comparison is appropriate when the channel is static. It is not when users walk: at 3 GHz and 1.5 m/s the channel stays coherent for about 28 ms, i.e., fewer than three 5G NR radio frames of 10 ms [12], whereas the optimizer with the highest $t = 0$ rate that we time needs about 5.9 s, roughly 600 frames. A method delivers its phase configuration to the channel that exists when it finishes, not to the channel it was handed. A slow method answers a question about a channel that is already gone.

This paper makes that observation quantitative for a multi-user MISO downlink assisted by a wall-mounted RIS in a 3GPP TR 38.901 indoor-office scenario [13] with pedestrian users ($v = 1.5$ m/s, $f_c = 3$ GHz, reference coherence time $T_c = 0.423/f_d = 28.2$ ms). Every phase is computed on the channel at $t = 0$ and evaluated on a geometrically evolved channel at the method's own measured latency. We show when, and by how much, latency changes which method one should use.

A. Contributions

1) **Effective-rate evaluation.** We define the effective rate of a method as its minimum user rate on the channel that exists at its own completion time (Section IV) and measure it for single-user heuristics, Adam, steepest ascent with Armijo line search, coordinate descent and a long-run Adam optimizer (1000 steps from 3 random starts). Between Adam and Armijo, too, the per-step cost decides which is better. The two methods with the highest rate at $t = 0$ need 0.9–5.9 s, and their effective rates, optimistic values of our channel model, fall below that of a 2.5 ms heuristic.
2) **An intrinsic optimum in the step count.** For Adam, the $t = 0$ rate keeps increasing with the number of steps L, but the effective rate peaks at $L = 4$–6 (5.98 b/s/Hz) and declines beyond. Running to $L = 48$ costs 14.0% (95% CI 12.5–15.5%). The optimum also exists for Armijo steepest ascent and for every Adam step size we tried, and it moves to fewer steps for faster users (8 steps at the 3 km/h and 3 at the 10 km/h of the indoor standards) and longer channel-acquisition delays, which we tie to the pilot overhead of cascaded channel estimation.
3) **A scaling law in the normalized step cost.** What sets the optimum is the step time relative to the channel's rate of change, $x = \Delta T f_d$. A small-delay model, approximately power-law convergence combined with quadratic aging, explains this as $L^\star \propto x^{-2/(\gamma+2)}$ (Proposition 1). The measured exponent, -0.675, lies within the model's range of -0.78 to -0.63, the closed form is within a factor of 1.3 of the measured optimum, and a 64-element surface has the same optima as the 32-element one. Sweeping x over two and a half decades moves

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the optimum from 96 steps to 2. The cost of running to near-convergence ($L = 256$) is 1.5% at the smallest x, peaks at about 21%, and falls again where little is left to gain. Because scaling the step time is equivalent to scaling the speed, this transfers the results to faster implementations and other speeds, approximately also to other carriers, and it states the condition under which convergence does *not* hurt.

4) **Periodic re-optimization.** In a controller that re-optimizes periodically and pays for the pilots of each update, the pilot duration sets the best update period (back-to-back to 80 ms), while a few steps per update remain best: with warm starts 2–8 steps, and warm starting gains up to 0.52 b/s/Hz.
5) **Ranking reversal between equal steps and equal milliseconds.** A learned (deep-unfolded) update rule beats a step-size-tuned Adam at every budget $L \in \{1, 2, 4, 8, 16\}$ when both run the same number of steps (win rates 68–84% of test scenarios). Measured in five independent timing rounds, every learned configuration is dominated by an Adam configuration that is both faster and better. One learned step costs 2.7–4.0 Adam steps. A published phase-aware CNN–LSTM [14], ported to a single-user version of the scenario, is dominated by a single, faster Adam step and does not improve coordinate ascent as its starting point.
6) **Pricing the modelling assumption.** All effective rates assume that the BS precoder is recomputed from the current channel while only the RIS phase ages. If the precoder aged together with the phase, the rate at T_c would be 11.7–14.2× lower; refreshing it costs about 13× less than one Adam step.

All tables and figures are generated by scripts from stored measurement files, and three measurement traps that produced wrong numbers in earlier drafts of this paper are described in the Appendix.

Section II reviews related work, Section III gives the system and channel model, Section IV defines the evaluation framework, Section V the measurement protocol, Section VI the results, and Section VII concludes.

II. Related Work

A. RIS Phase Optimization

Joint precoding and RIS phase design has been addressed with alternating optimization and semidefinite relaxation [2], beamforming under practical phase-dependent amplitude models [7], block-coordinate-descent designs for multicell systems [8], fractional programming for weighted sum-rate maximization [9] and fixed-point and manifold optimization [15], and WMMSE-based rate maximization [10] building on the WMMSE algorithm [16]. These methods are evaluated by the objective value they reach; their run time, when discussed, is characterized by complexity order rather than by measured latency relative to the channel's coherence time.

Learning-based designs replace the iterative solver with a trained policy, for example deep reinforcement learning for joint precoding and phase design in multi-user MISO systems [11], or learning to reflect and beamform directly from received pilots without explicit channel estimation [17]. Algorithm unrolling, which turns the iterations of an optimizer into the layers of a trainable network [18], is the template for the learned update rule studied here; it has been applied to IRS beamforming to avoid the cost of matrix inversions in an iterative solver [19]; it has also been used for MIMO detection [20] and for WMMSE precoding [21], [22]. Phase-aware regression, which predicts the sine and cosine of each phase instead of the phase itself, removes the $\pm\pi$ discontinuity that hampers direct regression of RIS phases [14]. In the learning-to-optimize literature a trained network approximates WMMSE at a fraction of its run time [23]. Learning-based designs are usually motivated by speed, yet that speed is rarely related to how fast the channel changes, and the classical solvers they are compared against are rarely tuned or timed under the same conditions.

B. Mobility and Outdated Channel Knowledge

High mobility has been addressed by mounting the surface on the vehicle so that a fast-fading channel becomes slow-fading [24]. The effect of channel aging [25] and of outdated channel state information [26] on RIS-aided systems has been analysed, and predictive beamforming, which acts on a forecast of the future channel, has been proposed for fast-moving links such as UAVs [27], and active sensing has been used to track mobile users with an RIS [28]. Channel aging has also been analysed for FDD massive MIMO [29], [30], and the cost of imperfect channel knowledge at the transmitter is classical in MIMO broadcast channels [31]. Two-timescale designs update the RIS phases from slowly varying channel statistics and the BS precoder from instantaneous channel knowledge [32], which reduces how often the phases must be recomputed; this has also been developed for fast-changing channels, where the instantaneous channel knowledge used by the precoder is itself quickly outdated [33]. In these works the delay between channel acquisition and use is a system parameter. The optimizer's own run time, which differs from method to method and from iteration count to iteration count, is not part of the model.

C. Discrete Phases

Practical surfaces have a few bits of phase resolution [34]; a recent prototype uses 2-bit unit cells with microcontroller-based control for fast reconfiguration of the whole array [35]. This paper uses continuous phases; the latency dimension of discrete phase design is left to future work (Section VII).

D. Gap Addressed Here

To our knowledge, no prior RIS work (i) evaluates each method on the channel at its own measured completion time, (ii) shows that this creates an interior optimum in the iteration count of a standard optimizer, or (iii) compares a learned update rule with a tuned classical optimizer both at equal steps and at equal measured milliseconds and finds that the two

comparisons rank them in opposite order. Run-time comparisons of learned and classical optimizers exist for interference management [23], but not, to our knowledge, for RIS phase design under mobility.

III. System and Channel Model

A. Signal Model

A BS with a uniform linear array of $N_t = 8$ antennas at half-wavelength spacing serves $K = 4$ single-antenna users through a direct link and an RIS of $N = 32$ passive elements (an 8×4 planar array, half-wavelength spacing); Section VI-C repeats the main sweep with an 8×8 surface ($N = 64$). With RIS phases $\boldsymbol{\theta} \in [0, 2\pi)^N$, $\mathbf{v} = e^{j\boldsymbol{\theta}}$, and precoder $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_K]$, user k receives

$$y_k = \big(\mathbf{h}_{d,k}^H + \mathbf{g}_k^H \,\mathrm{diag}(\mathbf{v})\mathbf{F}\big) \sum_{j=1}^{K} \mathbf{w}_j s_j + n_k, \qquad (1)$$

where $\mathbf{F} \in \mathbb{C}^{N \times N_t}$ is the BS–RIS channel, $\mathbf{g}_k^H \in \mathbb{C}^{1 \times N}$ the RIS–user channel, $\mathbf{h}_{d,k}^H \in \mathbb{C}^{1 \times N_t}$ the direct channel and $n_k \sim \mathcal{CN}(0, \sigma^2)$. Writing $\mathbf{h}_k^H(\boldsymbol{\theta})$ for the bracketed effective channel, the rate of user k is $R_k = \log_2(1 + \mathrm{SINR}_k)$ with

$$\mathrm{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \sigma^2}. \qquad (2)$$

For a given phase the precoder is regularized zero forcing [36] on the effective channel, with the MMSE regularizer $\lambda = K\sigma^2/P$ and total power P. The design objective is the minimum user rate, a fairness criterion also analysed for RIS-assisted MISO in [37],

$$\max_{\boldsymbol{\theta}} \; \min_k R_k(\boldsymbol{\theta}), \qquad (3)$$

which Adam and the learned update optimize through the smooth surrogate $-\frac{1}{\beta} \log \sum_k e^{-\beta R_k}$ with $\beta = 8$, evaluated with a log-sum-exp for numerical stability; coordinate descent and the long-run Adam reference maximize the hard minimum directly.

B. Geometry and Propagation

The room is $40 \times 20 \times 6$ m (Fig. 1). The RIS is mounted on a wall, centred at $(20, 0.1, 3)$ m and facing into the room; the BS is at $(20, 6, 4)$ m, in the panel's boresight and beyond its Fraunhofer distance. Users are dropped uniformly in the room at heights between 1 and 2 m. The carrier is $f_c = 3$ GHz, the bandwidth 20 MHz, the transmit power 20 dBm, the BS antenna gain 5 dBi and the receiver noise figure 7 dB.

All links follow the 3GPP TR 38.901 indoor-office model [13]: its path loss and LoS probability, a Rician factor K_R drawn from its distribution, and log-normal shadowing with Gudmundson spatial correlation [38] at a 5 m decorrelation distance (TR 38.901 specifies 10 m for LoS and 6 m for NLoS indoor links). Non-LoS components are built from point scatterers placed in the room and shared by all links of a scenario. The cascaded BS–RIS–user amplitude follows the two-hop path-loss model of Tang et al. [39] with a $\cos^q$ element pattern. Because an RIS is deployed where the direct path is obstructed, the direct link is attenuated by an additional 40 dB.

C. Mobility

Users walk in a straight line at $v = 1.5$ m/s (5.4 km/h) and are reflected at the walls. This lies between the 3 km/h of the 3GPP indoor-hotspot evaluation [40] and the 10 km/h upper end of the IMT-2020 pedestrian class, at which indoor mobility is evaluated [41]; Section VI-D repeats the main measurements at these speeds. Over the observation window the scatterer positions, per-path complex gains, LoS states, Ricean factors, shadowing values and the BS–RIS channel $\mathbf{F}$ are held fixed; only the user positions move, and the channels $\mathbf{g}_k(t)$ and $\mathbf{h}_{d,k}(t)$ are recomputed from geometry at every time instant. Channel variation therefore comes from geometry rather than from a stochastic aging model. As a reference time scale we use the Doppler frequency $f_d = v f_c / c = 15$ Hz and the coherence time $T_c = 0.423/f_d = 28.2$ ms [42], [43].

D. Known Modelling Limitations

We state the following limitations; the first two bias the rates in a known direction, while their effect on the optimal step count and on the rankings is not known.

- *Frozen scatterers.* Over seconds a walking user would see new scatterers, but the model keeps them. Over long horizons the modelled channel therefore changes less than a real one would, which *understates* the loss of slow methods. We treat latencies up to 200 ms as reliable and also report a stale floor, i.e., a $t = 0$ phase evaluated on an independent channel, as the true asymptote.
- *Independent per-antenna path gains.* The complex gains of the scattered paths are drawn independently for each BS antenna instead of being shared across the array. This adds antenna diversity, makes interference nulling easier, and so makes all rates optimistic.
- *Perfect channel knowledge.* Each method is given the exact channel at $t = 0$. Imperfect CSI is left to future work. The channel-acquisition delay is not part of the main results; Section VI-D adds it as a parameter.
- *Narrowband, far field.* The model is a single narrowband carrier, and for $N = 32$ the BS lies beyond the RIS's Fraunhofer distance (for $N = 64$ it lies just inside; the channel uses exact per-element distances). Wideband (OFDM) operation with beam squint and near-field operation of large surfaces are not covered.

IV. Latency-Aware Evaluation Framework

A. Effective Rate

Let a method A receive the channel $\mathcal{H}(0) = \{\mathbf{F}, \mathbf{g}_k(0), \mathbf{h}_{d,k}(0)\}$, return the phase $\boldsymbol{\theta}_A$, and need the wall-clock time τ_A to do so.

Definition 1 (Effective rate). The effective rate of A is its minimum user rate on the channel that exists when it finishes:

$$R_A^{\mathrm{eff}} = \min_k R_k\big(\boldsymbol{\theta}_A;\, \mathcal{H}(\tau_A)\big), \qquad (4)$$

where the precoder is recomputed from $\mathcal{H}(\tau_A)$.

The precoder is a closed-form digital computation that the BS can recompute whenever it has fresh effective-channel pilots, whereas the RIS phase requires the iterative optimization

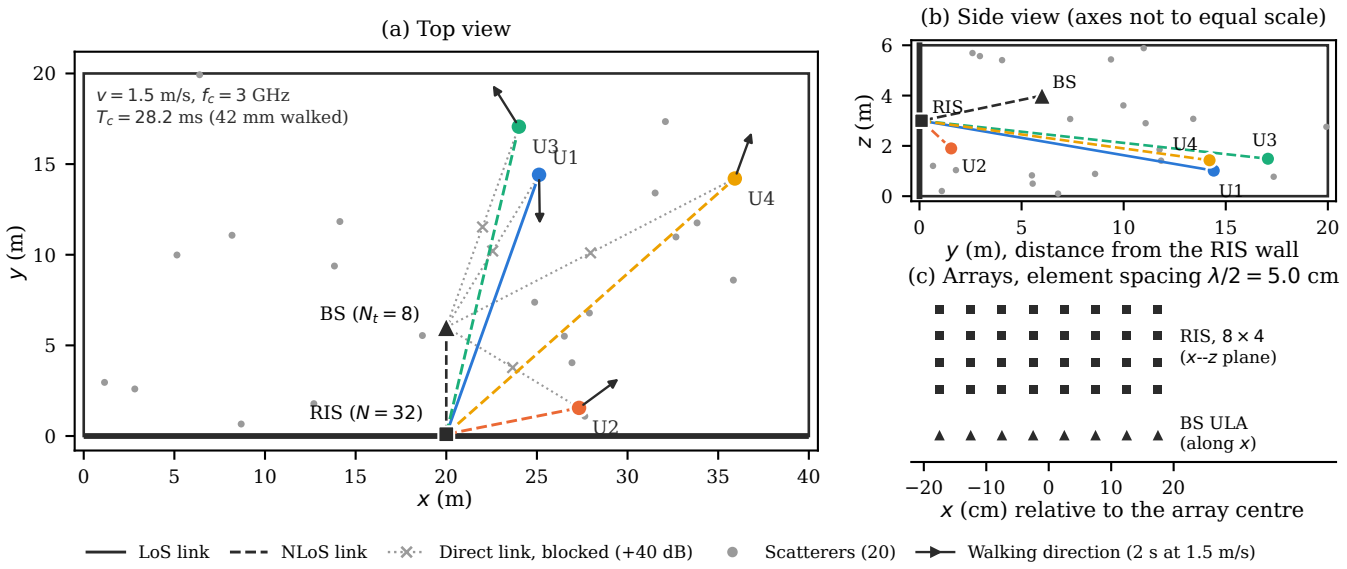


Fig. 1. The simulated scenario, drawn from the first scenario of the aging set (seed 11) exactly as the channel generator produced it. (a) Top view of the $40 \times 20 \times 6$ m office: the BS ($N_t = 8$) at $(20, 6, 4)$ m, the wall-mounted RIS ($N = 32$) centred at $(20, 0.1, 3)$ m, four users at 1–2 m height walking in straight lines at 1.5 m/s, and 20 point scatterers. Line style gives the LoS state drawn for each link from the 3GPP LoS probability. All direct links carry the 40 dB blockage. The arrows show 2 s of walking; within one coherence time a user moves only 42 mm. (b) Side view. (c) Element layouts of the RIS and the BS array.

whose latency is measured here. The same split between a fast precoder update and a slower phase update underlies two-timescale RIS designs [32]. Definition 1 therefore lets the phase age and keeps the beam fresh. Section VI-H measures how much this assumption is worth.

B. Gain–Aging Decomposition and the Intrinsic Optimum

For an iterative method run for L steps, write $r_0(L) = \min_k R_k(\boldsymbol{\theta}_L; \mathcal{H}(0))$ for its rate at $t = 0$, $\tau(L)$ for its latency and

$$R^{\text{eff}}(L) = r_0(L) - \Delta(L), \tag{5}$$

$$\Delta(L) = r_0(L) - \min_k R_k\big(\boldsymbol{\theta}_L; \mathcal{H}(\tau(L))\big), \tag{6}$$

so that $\Delta(L)$ is the rate lost to aging during the method's own run time. Additional steps raise r_0 with diminishing returns, while Δ grows with $\tau(L)$. An additional step pays off only while

$$r_0(L+1) - r_0(L) \;>\; \Delta(L+1) - \Delta(L), \tag{7}$$

and the effective rate peaks at the first L where this inequality fails. We call that point the *intrinsic optimum* $L^\star$. It is a property of the optimizer and its step size, the per-step latency and the speed of the channel together, not of the optimizer alone. A slower step or a faster user raises the right-hand side and moves $L^\star$ down. Equation (7) is a bookkeeping identity rather than a theorem: whether R^{eff} is unimodal is an empirical question, which we answer in Section VI-B.

C. Equal Steps versus Equal Milliseconds

Learned update rules are usually compared with classical optimizers at an equal number of steps. If one learned step costs c times as much as one classical step, an equal-step comparison gives the learned method c times the compute budget. We therefore also compare at equal measured time and use Pareto dominance.

Definition 2 (Dominance). Configuration B dominates configuration A if $\tau_B \le \tau_A$ and $r_{0,B} \ge r_{0,A}$, with one inequality strict, where r_0 is the rate at $t = 0$.

If B dominates A, there is no latency budget at which A is the better choice. A learned configuration that is dominated by some configuration of the classical optimizer is therefore never worth deploying, regardless of how it compares step for step.

D. Normalized Step Cost

Channel variation depends on how far a user moves in wavelengths, i.e., on $v\tau/\lambda = f_d\tau$. If an optimizer takes ΔT per step, its latency after L steps in these units is $L\,x$ with

$$x = \Delta T\, f_d. \tag{8}$$

In our model the users move $v\tau$ metres in time τ while everything else is fixed, so scaling the step time ΔT by a factor is exactly equivalent to scaling the speed v by the same factor at a fixed carrier; Section VI-C varies ΔT. The same x at a different carrier frequency is only approximately equivalent: at 28 GHz the user moves about 9 times fewer metres for the same x, which changes path loss, angles and the array's size in wavelengths.

E. A Small-Delay Model of the Optimum

Two approximate empirical regularities, examined in Section VI-C, make (7) explicit. First, over the step counts of

interest the gap of Adam's $t = 0$ rate to its converged value decays roughly as a power law,

$$R_\infty - r_0(L) \approx D\, L^{-\gamma}. \tag{9}$$

Second, for small delays the aging loss is quadratic in the distance moved, $u = f_d\tau$ wavelengths,

$$\Delta(L, u) \approx \kappa_L\, u^2, \tag{10}$$

A quadratic leading term is generic for a loss that is smooth in u and whose linear term cancels on average over directions of motion; for isotropic scattering it is the leading term of the Clarke correlation $J_0(2\pi u) \approx 1-\pi^2u^2$ [42], [43]. Our channel is geometric, so we fit κ_L rather than derive it. The coefficient grows with L: a more strongly optimized phase relies more on a coherent combination that motion destroys. With $\tau = L\Delta T$ we have $u = xL$, and the effective rate becomes

$$R^{\text{eff}}(L) \approx R_\infty - D\, L^{-\gamma} - \kappa\, x^2 L^2. \tag{11}$$

Proposition 1 (Scaling of the intrinsic optimum). Let $\gamma, D, \kappa, x > 0$. Under (9)–(10) with $\kappa_L = \kappa$, the model (11) has a unique maximizer over $L > 0$,

$$L^\star = \left(\frac{\gamma D}{2\kappa}\right)^{\frac{1}{\gamma+2}} x^{-\frac{2}{\gamma+2}}, \tag{12}$$

at which the aging loss equals $\gamma/2$ times the remaining convergence gap, $\kappa x^2 L^{\star 2} = \frac{\gamma}{2} D L^{\star\,-\gamma}$.

Proof. The derivative of (11) is $\gamma D L^{-\gamma-1} - 2\kappa x^2 L$. Since all constants are positive, it is positive for small L and negative for large L, and vanishes only where $L^{\gamma+2} = \gamma D/(2\kappa x^2)$, which gives (12). The second derivative, $-\gamma(\gamma+1)DL^{-\gamma-2} - 2\kappa x^2$, is negative, so the stationary point is the maximum. Substituting (12) into $\kappa x^2 L^2$ gives the stated balance. □

Two qualitative consequences follow. The optimum is a power law in x, and because $\gamma > 0$ its exponent lies in $(-1, 0)$: a faster optimizer should run more steps, but less than proportionally more. The balance identity is a property of the model; Section VI-C shows that the data do not follow it quantitatively, which marks where the model stops being accurate. The derivation uses only small-delay aging and so requires $u^\star = xL^\star \ll 1$, which holds over the range we test ($u^\star \le 0.16$). With fitted constants, (12) is a description of the measurements rather than an independent prediction; its value lies in explaining why the optimum shifts as it does.

F. Periodic Re-optimization

Definition 1 evaluates a phase at the instant it is delivered. A deployed controller instead repeats the optimization with period U: at each snapshot it spends T_{p} on pilots for the cascaded channel, during which no data is transmitted, then $L\Delta T$ on optimization, deploys the result, and waits for the next snapshot ($U \ge T_{\text{p}} + L\Delta T$; *back-to-back* operation means $U = T_{\text{p}} + L\Delta T$).

Definition 3 (Time-averaged effective rate). Let $\boldsymbol{\theta}(t)$ be the phase deployed at time t, and let $\mathcal{P}$ be the set of pilot intervals. The time-averaged effective rate over a window $\mathcal{W}$ is

$$\bar{R}^{\text{eff}} = \frac{1}{|\mathcal{W}|} \int_{\mathcal{W}\setminus\mathcal{P}} \min_k R_k\big(\boldsymbol{\theta}(t);\, \mathcal{H}(t)\big)\, dt. \tag{13}$$

Each update can start from scratch (*cold start*) or from the phase currently deployed (*warm start*). The period trades pilot overhead against the age of the phase in use, and L trades the quality of each update against its delay.

V. Measurement Protocol

A. Scenario Sets

We use five scenario sets, all generated with seed 11, and never mix numbers from different sets in one table or comparison; Section VI-G additionally uses a training draw (seed 2026) and a selection draw (seed 12), described below. Absolute rates differ between the sets, mainly because their numbers of scatterers differ.

- *Aging set* (Tables I, III and V, Figs. 2–4): 400 scenarios with walking users and 20 point scatterers per scenario, close to the 15–19 clusters of the 3GPP indoor-office model. The periodic re-optimization study regenerates the same 400 scenarios on a finer time grid; the generator's random draws do not depend on the time grid, and the SU-average rate at 0, 50, 100 and 200 ms reproduces the aging run.
- *Aging set with* $N = 64$ (Table II, Fig. 4(a)): the same construction with an 8×8 panel. Its Adam step time is the 1.66 ms of $N = 32$ scaled by the measured ratio of the two step times, 1.04 (one thread, 20 scenarios, 7 repetitions each, one round; load average 0.40).
- *Beam set* (Table IX): a separate draw of 400 scenarios with walking users and the generator's default of 6 scatterers.
- *Single-user set* (Table VIII): the aging set's construction with one user, 400 scenarios. The CNN–LSTM of Section VI-G is trained on 28 000 of 40 000 static scenarios of the same generator drawn with seed 2026; 6000 select its checkpoint and Adam's step size, and 6000 form a static test set. None of these is an evaluation scenario. The best method at each T_{ov} is selected on a further 400 scenarios drawn with seed 12, and the $t = 0$ comparison is repeated on two static draws of 6000 scenarios (seeds 11 and 13).
- *Static set* (Tables VI and VII, Fig. 5): 6000 scenarios without mobility, split 70/15/15 into 4200 training scenarios and two pools of 900, from which 400 validation and 600 test scenarios are used. The generator is not prefix-stable, i.e., drawing n scenarios is not the same as taking the first n of a larger draw, so every trained model is evaluated with the sample count it was trained with.

B. Methods

SU-average and *SU-best-of* are closed-form single-user phase alignments; the latter keeps the best of $K+1$ candidates. *Adam* [44] optimizes the soft-min surrogate starting from SU-average. *Armijo* is steepest ascent with Armijo's sufficient-increase rule [45] on the same surrogate and from the same start, with a per-scenario backtracking line search: sufficient-increase constant 10^{-4}, halving, the step doubled after each accepted iteration and reset to the initial step a_0 after a failed search. *Coordinate descent* searches each element over

a 16-level phase grid with the others fixed, for up to 8 sweeps. *Long-run Adam* is Adam with step size 0.05 for 1000 iterations from 3 random starts, keeping the best; both it and coordinate descent maximize the hard minimum. The *learned update* is a deep-unfolded network with L learned steps. Each step receives the gradient of the soft-min rate as input and exchanges information across users. A separate network is trained for each L, without labels, with the soft-min rate as the training objective.

Hyper-parameters on the static set. At every budget $L \leq 16$, Adam's step size is selected on the validation set from ten values in $[0.02, 2.0]$, together with the objective (soft-min or hard min). The selected step size is 0.75 with the soft-min objective at every L, an interior point of the grid. The larger budgets used only in the latency comparison ($L = 24, 32, 48, 64$) are tuned the same way on six values in $[0.2, 1.5]$. Comparing a learned rule with an untuned optimizer would overstate its advantage.

Hyper-parameters on the aging set. Adam uses step size 0.5; repeating the sweep with 0.25, 0.75 and 1.0 shows that 0.5 gives the highest peak. Armijo's initial step a_0 is chosen from $\{0.1, 0.3, 1, 3, 10\}$ by its best effective rate at Adam's per-step latency, before Armijo's own cost is measured (3 is selected). Both choices use the same 400 scenarios on which results are reported, so they are robustness checks rather than held-out selections.

C. Timing

All latencies are wall-clock times for a single scenario on an Intel Xeon CPU at 2.1 GHz with one PyTorch thread, measured with `time.perf_counter`; the median over repetitions is kept, and most measurements are repeated in several rounds. These are latencies of an unoptimized Python implementation; Section VI-C shows how the results change for faster implementations.

- *Static set* (Table VII). Adam and the learned update are timed one scenario at a time on a separate 64-scenario draw of the static generator, from a shared, precomputed SU-average start, 25 repetitions per configuration, five rounds; the rates come from the test set. The learned network gets three warm-up calls. Its input features are precomputed and excluded from the timing, which favours the learned method. The round-to-round spread is 1.15–1.60× (max/min). The timing script does not record the load average.
- *Aging set* (Table I). SU-average, SU-best-of, coordinate descent, long-run Adam and Adam with $L \in \{1, 2, 4, 8, 16, 32, 48\}$ are timed with one script, five rounds. The load average was 0.20 before the run and 0.86 at its end, which is what the script's own busy thread produces; the round-to-round spread is 1.15–3.66×, largest for the sub-millisecond SU-average. The Adam rows and all step sweeps use a per-step latency of 13.3/8 = 1.66 ms from an earlier, two-thread measurement made before a precoder fix, which lies within the range measured here (1.49–2.02 ms per step; least-squares slope 1.50 ms).
- *Single-user set* (Table VIII). Every method, including the network's feature construction, is timed on 20 scenarios, 25 repetitions each, three rounds; the script waits until the load average is below 0.5; timing began at 0.16 and ended at 0.97. Adam's step size is chosen at each L from ten values in $[0.02, 2.0]$ on 400 validation scenarios of the training draw.
- *Armijo versus Adam.* Armijo's cost per step depends on the number of backtracking evaluations, so both are timed for 6 steps (Armijo's optimum at Adam's step time) on 20 different scenarios. The ratio of the medians is 1.26 (0.81–1.97 across scenarios), giving 2.09 ms per Armijo step. The load average was 0.18 when the script started and 0.83 when the timing began, after the script's own computations. Because the ratio varies between scenarios, Section VI-B reports how the comparison depends on it.

D. Aging Sweeps and Statistics

The sweeps evaluate each phase on a fixed time grid and interpolate linearly at the method's latency plus the acquisition delay T_{ov}. Between grid points, and especially at 3 m/s where the grid is coarse relative to T_c, the effective rates are interpolated, and differences of a few thousandths of a b/s/Hz between neighbouring L are not resolved. Latencies beyond the last grid point (5.6 s) take its value.

Absolute rates vary strongly between scenarios: the 95% confidence intervals of the scenario means in Table I are ±0.17–0.24 b/s/Hz. Comparisons between methods or step counts are therefore made with a *paired* bootstrap [46] over the 400 scenarios (2000 resamples), and we call a configuration *not significantly below the best* if its paired difference to the best is not significantly negative at the 95% level. Table III instead lists the step counts *within 0.5%* of the best, a simpler criterion. Best configurations are selected and tested on the same scenarios without a multiple-comparison correction; they should be read as estimates, with the stated ranges as their uncertainty.

E. Acquisition Delay and Pilot Overhead

We tie T_{ov} and T_{p} to the pilot overhead of cascaded channel estimation. Cascaded-channel estimation for RIS is surveyed in [47]. With the scheme of [48], $K + N + \max(K - 1, \lceil (K-1)N/N_t \rceil) = 48$ pilot symbols suffice for $K = 4$, $N = 32$, $N_t = 8$; estimating each user separately with $N + 1$ reflection patterns would take $K(N+1) = 132$. With 5G NR numerology (14 OFDM symbols per slot of $15/\mathrm{SCS}$ ms [12]) and one OFDM symbol per pilot, 48 symbols last 3.43, 1.71 and 0.86 ms at 15, 30 and 60 kHz subcarrier spacing, and 132 symbols last 9.43, 4.71 and 2.36 ms. Feedback, estimation processing and the switching time of the RIS come on top and are not modelled. The range $T_{\mathrm{ov}} \in [0, 10]$ ms of Table III covers these cases, and Table V uses $T_{\mathrm{p}} \in \{0, 0.86, 1.71, 3.43\}$ ms.

F. Periodic Re-optimization

The controller of Definition 3 is simulated on a 0.5 ms grid over 200 ms, the horizon within which the mobility model

is reliable. No data is transmitted during pilot intervals. We test update periods $U \in \{10, 20, 40, 80\}$ ms and back-to-back operation, with $L \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$ steps per update wherever $T_\mathrm{p} + L\Delta T \leq U$. The rate is averaged over 100–200 ms, after every policy has deployed at least once; the phase in use before the first deployment is SU-average.

G. Reproducibility

One script generates every table and figure in this paper from the stored result files. Re-running the aging sweeps reproduces the stored results bit for bit, and the newer experiments (aging statistics, step scaling, the small-delay model and periodic re-optimization) assert that they reproduce the stored means of the aging run before reporting anything. The project's separate audit script checks 98 assertions about the learning experiments of the underlying thesis; among the numbers reported here it covers the verdicts of Section VI-G and the standardized-speed results of Section VI-D.

VI. Results

A. Aging of a Fixed Phase and Effective Rates

In this scenario the RIS carries most of the signal. With the direct links blocked, the median per-user SNR (channel gain over all BS antennas times P/σ^2) is 20.0 dB for the direct link alone and 31.1 dB for the RIS path with the SU-average phase. Without the RIS, the minimum user rate is 1.744 b/s/Hz; a random phase gives 4.072 (Table I).

Fig. 2 fixes each method's phase at $t = 0$ and tracks the minimum user rate as users walk, with the precoder refreshed at every instant. Rates are flat for the first few milliseconds, drop sharply between roughly 5 and 30 ms, i.e., around T_c, and level off above the stale floor. The more a phase has been optimized, the more it has to lose: by $\tau = T_c$ long-run Adam has fallen from 7.117 to 5.473 b/s/Hz and Adam with $L = 8$ from 6.443 to 5.165, while SU-average has only moved from 4.624 to 4.503.

Table I evaluates each method at its own measured latency (Definition 1). Long-run Adam and coordinate descent have the best $t = 0$ rates but need 5.9 s and 0.88 s, by which time users have moved metres. Their effective rates lie beyond the 200 ms horizon where our model is reliable and are optimistic; the true values lie between the tabulated effective rate and the stale floor. Even the optimistic values are below that of SU-best-of, which takes 2.5 ms: in a paired comparison SU-best-of is ahead of coordinate descent by 0.297 b/s/Hz (95% CI 0.204–0.390) and of long-run Adam by 0.604 (0.415–0.805). Among the methods in Table I, Adam with $L = 8$ has the highest effective rate (5.849 b/s/Hz); the step sweep below shows that fewer steps do better still.

B. The Intrinsic Optimum

Fig. 3 sweeps the number of optimizer steps, starting from SU-average (4.624 b/s/Hz); the point at $L = 0$ shows SU-best-of for reference. Adam's $t = 0$ rate grows monotonically with L, from 5.412 at $L = 1$ to 7.037 b/s/Hz at $L = 48$. The effective rate does not. It has a flat top between $L = 4$ and $L = 6$ (5.980, 5.982 and 5.970 b/s/Hz, at 6.7–10.0 ms); none of the three is significantly below the best, and every other step count is. At $L = 48$ (79.8 ms) the effective rate is 5.145 b/s/Hz, 14.0% (95% CI 12.5–15.5%) below the optimum, although the $t = 0$ rate there is 13.2% higher.

In terms of (7): going from $L = 4$ to $L = 6$ adds 0.188 b/s/Hz at $t = 0$, but the additional 3.3 ms of aging cost 0.198 b/s/Hz. Between $L = 6$ and $L = 48$ the effective rate declines; the only exception is a 0.006 b/s/Hz fluctuation between $L = 20$ and $L = 24$.

The optimum is not specific to Adam or to its step size. Repeating the Adam sweep with step sizes 0.25, 0.75 and 1.0 moves the best point to $L = 6$, 4 and 5, and running to $L = 48$ costs 11.5%, 13.7% and 11.3%. Armijo steepest ascent has its best effective rate at $L = 4$ (6.015 b/s/Hz at 2.09 ms per step). It gains more than Adam in the first steps (6.232 against 6.128 b/s/Hz at $L = 4$) but costs 1.26 times as much per step, and the two best effective rates are

TABLE I

Rate at $t = 0$, latency, and effective rate on the aging set (400 scenarios, $v = 1.5$ m/s). Stale floor: $t = 0$ phase on an independent channel. Adam rows use 1.66 ms per step; other latencies are measured (Section V). 95% CIs of these scenario means are ± 0.17–0.24 b/s/Hz; comparisons in the text use paired differences. †Beyond the 200 ms reliable horizon; optimistic. *Beyond the last time-grid point (5.6 s); value at 5.6 s.

Method	t=0 (b/s/Hz)	Latency (ms)	Effective (b/s/Hz)	Stale (b/s/Hz)
No RIS (direct link)	1.744	–	–	–
Random phase	4.072	–	–	–
SU-average	4.624	0.42	4.623	4.315
SU-best-of	5.276	2.5	5.272	4.336
Adam, $L = 8$	6.443	13.3	5.849	4.326
Adam, $L = 12$	6.631	20.0	5.515	4.370
Adam, $L = 16$	6.754	26.6	5.324	4.381
Adam, $L = 20$	6.835	33.2	5.276	4.364
Coord. descent	6.925	880.8	4.975†	4.352
Long-run Adam	7.117	5851.7	4.667$^{\dagger *}$	4.423

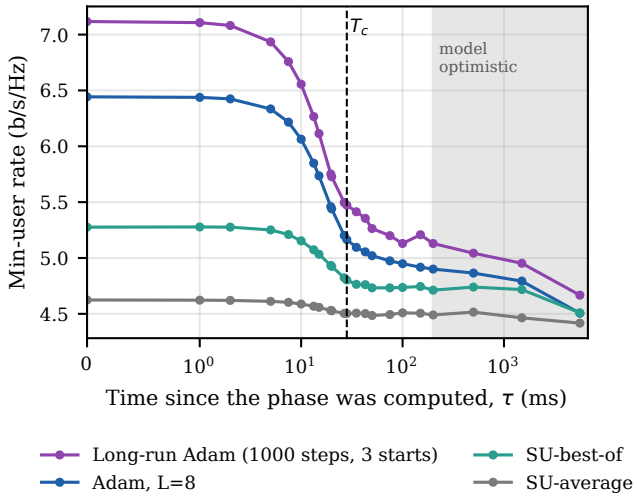


Fig. 2. Minimum user rate of a phase fixed at $t = 0$ as users walk at 1.5 m/s, with the precoder refreshed at every instant. Dashed line: $T_c = 28.2$ ms. Shaded: beyond 200 ms the frozen-scatterer model is optimistic.

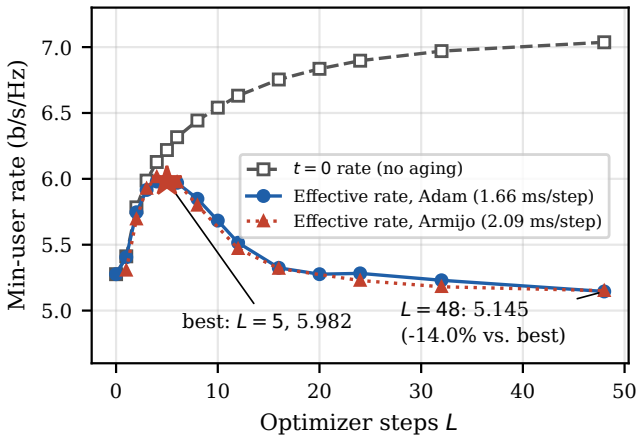


Fig. 3. Step sweep on the aging set. Adam's $t = 0$ rate increases monotonically, while the effective rate (rate at the method's own completion time) has a flat top at $L = 4$–6 and is 14.0% below its best value at $L = 48$. Armijo steepest ascent (2.09 ms per step) shows the same shape. $L = 0$: SU-best-of, for reference, evaluated at 0.05 ms (at its measured 2.5 ms the effective rate is 5.272 instead of 5.276).

statistically indistinguishable: Armijo is ahead by 0.033 b/s/Hz (95% CI -0.001 to 0.064). Which of the two is better depends on the cost ratio. With Armijo's per-step cost set to 1.0, 1.2, 1.4, 1.6 and 2.0 times Adam's, its best effective rate is $+0.147$, $+0.050$, -0.013, -0.079 and -0.197 b/s/Hz relative to Adam's. Even between two classical optimizers, the per-step latency decides the ranking.

C. Dependence on the Step Cost

The latencies above are those of a single-threaded Python implementation; an optimized implementation would be much faster per step. Fig. 4 therefore scales Adam's per-step latency from 0.01 to 3 times the measured value, which moves the normalized step cost $x = \Delta T f_d$ of (8) from 2.5×10^{-4} to 7.5×10^{-2}; this setup sits at $x = 2.5\times10^{-2}$. By the equivalence of Section IV, the same axis describes users who are up to 100 times slower or 3 times faster.

The best step count falls steadily with x: 96 steps at $x = 2.5\times10^{-4}$, 48 at 7.5×10^{-4}, 24–32 at 2.5×10^{-3}, 12–16 at 7.5×10^{-3}, 4–6 here, and 2 at $x = 7.5\times10^{-2}$. Adam's $t = 0$ rate saturates near 100 steps (7.087 b/s/Hz at $L = 96$, 7.096 at $L = 256$), so at the smallest x the optimum is essentially convergence, and running to $L = 256$ costs only 1.5% (95% CI 1.3–1.8%). The cost of convergence peaks at about 21% for $x \approx 2.5\times10^{-3}$, where the optimizer is fast enough to gain much but convergence still takes long. It declines again for large x, where even the best policy has little left to gain. For $x \geq 2.5\times10^{-2}$, $L = 256$ lies beyond the 200 ms horizon, so its cost is if anything understated.

The claim that convergence hurts is thus a statement about a regime, not about RIS optimization in general. With a hundred-fold faster implementation at pedestrian speed, convergence is nearly free. The same implementation at vehicular speed, or at a millimetre-wave carrier, moves x in the opposite direction.

Model versus measurement. Both laws behind Proposition 1 hold approximately on the aging set. The aging loss is close to quadratic, $R^2 \geq 0.98$ for every L (fitted for $\tau \leq 10$ ms), and κ_L grows from 7.8 at $L = 1$ to 25.1 at $L = 48$, roughly as $L^{0.24}$. The convergence gap is only approximately a power law: fitted over $1 \leq L \leq 64$ with R_∞ taken at $L = 256$, it gives $\gamma = 0.93$ ($R^2 = 0.90$ on a log scale), but the residuals are systematic, and the fitted γ ranges from 0.56 to 1.19 as the upper end of the fit range moves from 16 to 128 steps. The predicted exponent $-2/(\gamma+2)$ correspondingly ranges from -0.78 to -0.63; letting κ_L grow as $L^{0.24}$ in the same derivation gives -0.63. The measured slope of $L^\star(x)$ is -0.675 (95% bootstrap CI -0.692 to -0.655), inside that range (Table II, Fig. 4(a)). The data are therefore consistent with a power law of the predicted form, but the model pins the exponent down only to about ±0.08.

With κ set to the median of κ_L over $2 \leq L \leq 48$, the closed form (12) is within a factor of 1.27 of the measured optimum at every x; at this setup it gives $L^\star = 5.3$ against a measured 5. Keeping the measured $r_0(L)$ and κ_L and maximizing (11) over the step grid gives the measured optimum in 10 of 12 cases (both surfaces, six values of x), and in the other two a step count within 0.14% of the best. This is an in-sample consistency check: κ_L is fitted over the same delays (1.6–10 ms) at which the optima occur. The balance of Proposition 1 does not hold quantitatively: at the measured optimum the ratio of aging loss to remaining gap varies from 0.19 to 1.65 across x instead of staying at $\gamma/2 = 0.47$, because κ_L grows with L and the gap is not an exact power law. The model thus explains where the optimum lies and why it moves as a power of x, with three constants fitted to the convergence and aging curves; it does not describe how the two losses split.

A larger surface. We repeat the sweep with an 8×8 RIS ($N = 64$) in the same room, with the same BS, users, scatterers and seed. The larger surface raises the converged rate ($R_\infty = 8.48$ against 7.10 b/s/Hz), but the best step count is identical to that of $N = 32$ at every step-time multiplier: $L^\star = 5$ at this setup, and running to $L = 48$ costs 14.8% (point estimate). Because the optimal step counts coincide, the slope of $L^\star(x)$ is the same by construction, so the larger surface confirms the optimum, not the exponent independently. The fitted constants differ ($\bar\kappa = 22.0$ against 18.8), and the closed form is again within a factor of 1.27. In our implementation an Adam step costs only 1.04 times more at $N = 64$ than at $N = 32$, which suggests that fixed per-call overhead rather than arithmetic dominates the step time (we did not profile this); an implementation whose cost grows with N would place the larger surface at a larger x and, by (12), at a smaller $L^\star$. For $N = 64$ the BS (6.0 m from the panel centre) lies just inside the Fraunhofer distance of the aperture diagonal (6.4 m); the channel uses exact per-element distances, so this does not affect the results, but the far-field statement of Section III applies to $N = 32$.

D. Sensitivity to Acquisition Delay and User Speed

The effective rate above charges each method only for its own run time. A deployed system also spends time on pilots, estimation and feedback before the optimizer starts, which for this system is between about 1 and 10 ms (Section V). We

TABLE II
SMALL-DELAY MODEL (PROPOSITION 1) AGAINST MEASUREMENT, FOR TWO RIS SIZES ON THE AGING-SET GEOMETRY. THE PREDICTED EXPONENT USES γ FITTED OVER $1 \le L \le 64$; OTHER FIT RANGES GIVE -0.78 TO -0.63. THE MEASURED EXPONENTS COINCIDE BECAUSE THE OPTIMAL STEP COUNTS DO.

	$N = 32$	$N = 64$
Convergence exponent γ	0.93	0.93
Aging coefficient $\bar{\kappa}$	18.8	22.0
Predicted exponent $-2/(\gamma+2)$	-0.682	-0.683
Measured exponent of $L^\star(x)$	-0.675	-0.675
$L^\star$ at this setup (measured)	5	5
$L^\star$, closed form (12)	5.3	5.2
Loss of L=48 vs. $L^\star$	14.0%	14.8%

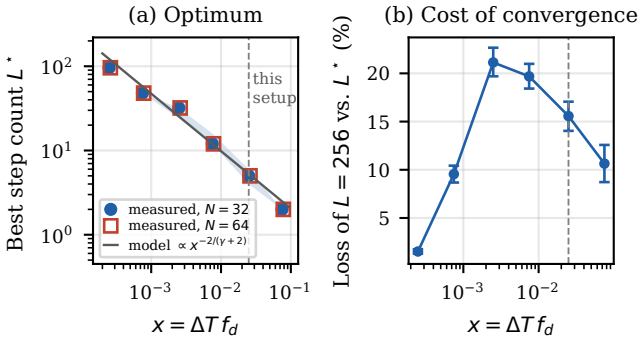


Fig. 4. Adam on the aging set with its per-step latency scaled from 0.01 to 3 times the measured value. (a) Best step count for $N = 32$ (shaded: step counts not significantly below the best) and $N = 64$; line: closed form (12) with the $N = 32$ constants. Panel (b) and the shading refer to $N = 32$. (b) Effective-rate loss of running to $L = 256$ instead of $L^\star$, with 95% paired bootstrap CIs. Dashed: this setup.

therefore add an acquisition delay T_{ov} to every latency and repeat the sweep at four user speeds (Table III). The direction is the one predicted by (7): a faster channel or a longer delay makes each additional step cost more aging, and the optimum moves to fewer steps. Without delay, the range of L within 0.5% of the best shifts from $L = 6$–8 at 1 m/s to $L = 3$ at 3 m/s, and at 1.5 m/s it shifts from $L = 4$–6 to $L = 3$–4 as T_{ov} grows from 0 to 10 ms. The best effective rate falls with T_{ov} at every speed. We do not interpret cells with $T_{\text{ov}} > T_c/2$, where the channel has largely changed before optimization starts. The only such cell is 3 m/s with 10 ms; there all step counts give rates within 0.26 b/s/Hz, and the maximum falls at $L = 32$, which carries no meaning. In all other cells, running Adam to $L = 48$ costs 2.9–14.7% of the attainable effective rate.

The optimum also tolerates an error in the assumed speed: choosing L for 1.5 m/s when users move at 1.8 m/s, or the reverse, costs less than 0.5%, and at most 5.7% (chosen for 1 m/s, users at 3 m/s).

Standardized speeds. Table IV repeats the main measurements on the same 400 scenarios, on a 0.5 ms grid up to 40 ms, at the speeds of the standards: 3 km/h, used in the 3GPP indoor-hotspot evaluation [40]; 10 km/h, the upper end of the IMT-2020 pedestrian class at which indoor mobility is evaluated [41]; and, as a stress test outside the indoor classes, the 30 km/h of the dense-urban evaluation. With $T_{\text{ov}} = 0$, both findings hold at every speed. The optimum moves from 8 steps at 3 km/h to 3 at 10 km/h and 1 at 30 km/h, and running to $L = 48$ costs 14.5% (95% CI 13.1–16.1%) at 3 km/h and 10.5% (9.0–11.9%) at 10 km/h. SU-best-of is ahead of coordinate descent by 0.25–0.42 b/s/Hz and of long-run Adam by 0.45–0.64 at every speed, with paired CIs above zero. At 30 km/h, $L \ge 24$ lies beyond the reliable horizon (36 ms), so the $L = 48$ cost is approximate, and with an acquisition delay nothing is resolved. At 1.5 m/s the finer grid makes $L = 4$ and 5 tie ($L^\star = 4$, 14.2%; main sweep 5, 14.0%).

TABLE III
BEST NUMBER OF ADAM STEPS $L^\star$ ON THE GRID $L \in \{1, \dots, 6, 8, 10, 12, 16, 20, 24, 32, 48\}$ AND, IN PARENTHESES, THE RANGE OF L WITHIN 0.5% OF THE BEST EFFECTIVE RATE, FOR DIFFERENT USER SPEEDS AND CHANNEL-ACQUISITION DELAYS T_{ov} (AGING SET). $^\ddagger T_{\text{ov}} > T_c/2$: NOT INTERPRETED.

	Acquisition delay T_{ov} (ms)				
v (m/s), T_c	0	1	2	5	10
1, 42.3 ms	6 (6–8)	6 (6–8)	6 (5–8)	6 (5–6)	5 (4–6)
1.5, 28.2 ms	5 (4–6)	4 (4–5)	4 (4–5)	4 (3–4)	3 (3–4)
1.8, 23.5 ms	4 (4–5)	4 (3–4)	4 (3–4)	3 (3–4)	3 (3–4)
3, 14.1 ms	3 (3)	3 (3)	3 (2–3)	2 (2–3)	$32^\ddagger$ (32–48)

TABLE IV
THE MAIN MEASUREMENTS AT STANDARDIZED USER SPEEDS (AGING-SET SCENARIOS, $T_{\text{ov}} = 0$): COHERENCE TIME, NORMALIZED STEP COST, BEST STEP COUNT (RANGE WITHIN 0.5%), COST OF RUNNING ADAM TO $L = 48$, AND Δ: SU-BEST-OF MINUS COORDINATE DESCENT IN EFFECTIVE RATE (B/S/HZ). AT 30 KM/H, $L = 48$ LIES BEYOND THE RELIABLE HORIZON.

km/h	Source	T_c (ms)	x	$L^\star$	L=48	Δ
3	3GPP [40]	50.8	0.014	8 (6–8)	14.5%	+0.27
5.4	ours	28.2	0.025	4 (4–6)	14.2%	+0.31
10	ITU [41]	15.2	0.046	3	10.5%	+0.25
30	ITU, urban	5.1	0.139	1	4.3%	+0.42

E. Periodic Re-optimization

Table V simulates the controller of Definition 3 and reports, for each pilot duration, the best combination of update period U and steps per update L. Pilot overhead is what sets the period. If pilots were free, back-to-back updates would be best, with two warm-started steps (6.36 b/s/Hz) or three cold-started ones. With 0.86 ms of pilots per update, the best period is 10–20 ms; with 1.71 ms it is 20 ms for warm starts; with 3.43 ms the best policies update every 80 ms, or back-to-back with 48 steps, which amounts to the same thing. Short back-to-back cycles then spend most of their time on pilots.

What does not change is that few steps per update suffice. With warm starts the best policy uses 2, 3, 3 and 8 steps for the four pilot durations. In every row, some policy with at most 8 steps per update is not significantly below the best one, and with warm starts and pilots of up to 1.71 ms the best policy is significantly better than every policy with more steps. Warm starting gains 0.52, 0.22 and 0.15 b/s/Hz for the three shorter pilot durations; with 3.43 ms pilots updates are so rare that the gain vanishes.

TABLE V
PERIODIC RE-OPTIMIZATION (DEFINITION 3, AGING SET): BEST UPDATE PERIOD U (MS; B2B = BACK-TO-BACK) AND STEPS PER UPDATE L, WITH THE TIME-AVERAGED EFFECTIVE RATE OVER 100–200 MS (B/S/HZ), FOR PILOT DURATIONS T_{p} WITHOUT DATA TRANSMISSION. WARM−COLD: PAIRED DIFFERENCE OF THE TWO BEST POLICIES (B/S/HZ); §NOT SIGNIFICANT AT 95%.

	Cold start			Warm start			
T_{p} (ms)	U	L	rate	U	L	rate	warm−cold
0	b2b	3	5.83	b2b	2	6.36	+0.52
0.86	20	4	5.20	10	3	5.42	+0.22
1.71	b2b	48	5.09	20	3	5.23	+0.15
3.43	b2b	48	4.98	80	8	4.99	+0.01§

TABLE VI
EQUAL NUMBER OF STEPS, STATIC TEST SET (600 SCENARIOS). ADAM STEP SIZE AND OBJECTIVE SELECTED ON 400 VALIDATION SCENARIOS AT EACH L.

L	Learned (b/s/Hz)	Adam (b/s/Hz)	Ratio	Diff. -2 s.e. (b/s/Hz)	Wins (%)
1	4.460	4.381	1.018	+0.061	74.8
2	4.879	4.693	1.040	+0.160	83.2
4	5.285	5.025	1.052	+0.228	83.7
8	5.593	5.380	1.040	+0.184	83.2
16	5.800	5.739	1.011	+0.039	68.3

In short, periodic operation confirms the single-update result: the channel-acquisition cost decides how often to update, and the effective-rate trade-off decides how much to compute per update, which is little.

F. Learned versus Classical: Equal Steps and Equal Milliseconds

This comparison is made on the static set, without mobility, and with $t = 0$ rates. It isolates the question of compute efficiency, which the previous subsections show to be what matters once latency enters. The learned model was not evaluated on the aging set.

Table VI compares the learned update rule with tuned Adam at equal step counts on the static test set; both start from SU-average (3.539 b/s/Hz). The learned rule is ahead at every budget, the paired difference minus two standard errors is positive at every budget, and it wins on 68–84% of test scenarios. On a step axis, the learned rule is the better optimizer.

Steps, however, are not equally expensive. One learned step takes 4.6–6.5 ms, one Adam step 1.60–1.71 ms, so a learned step costs 2.7–4.0 Adam steps. Table VII and Fig. 5(b) place both methods on the measured latency axis. Every learned configuration is dominated by an Adam configuration that is both faster and better. This holds in each of the five timing rounds separately, although not always by the configuration listed: for learned $L = 16$, Adam with $L = 48$ is slower in two rounds, and Adam with $L = 32$ (6.013 b/s/Hz, at most 62.97 ms) dominates in every round. For example, the learned rule with $L = 4$ needs 18.41 ms for 5.285 b/s/Hz, while Adam with $L = 8$ needs 13.48 ms for 5.380 b/s/Hz. The ranking from the equal-step comparison reverses completely.

TABLE VII
EQUAL MILLISECONDS: EACH LEARNED CONFIGURATION AND THE BEST ADAM CONFIGURATION THAT DOMINATES IT (LOWER MEDIAN LATENCY AND HIGHER RATE). MEDIANS OVER FIVE ROUNDS. IN EVERY ROUND EACH LEARNED CONFIGURATION IS DOMINATED BY AT LEAST ONE ADAM CONFIGURATION.

Learned update			Dominating Adam		
L	ms	b/s/Hz	L	ms	b/s/Hz
1	5.16	4.460	2	3.20	4.693
2	12.95	4.879	4	6.49	5.025
4	18.41	5.285	8	13.48	5.380
8	36.76	5.593	16	27.31	5.739
16	83.80	5.800	48	78.77	6.107

Three remarks bound this result. First, aging was not included; since the learned configuration is the slower one in every pair, aging would be expected to widen rather than close the gap, but we have not measured this. Second, the learned timing excludes feature computation, so the latency gap is if anything understated. Third, the cost ratio is a property of this implementation: on hardware that accelerates the network's matrix products more than Adam's step, the reversal could weaken.

G. A Published Learned Phase Predictor under Mobility

The learned update rule above is our own design. We also test a learned phase predictor published earlier by the authors for a different setting, the phase-aware CNN–LSTM of [14]. It predicts $(\sin\theta_n, \cos\theta_n)$ for every element, which avoids the $\pm\pi$ discontinuity of direct phase regression, and it was proposed for a single-antenna BS, where the optimal phase of each element has a closed form. We keep the architecture of its released MATLAB code (two convolutional blocks, two LSTM layers over the elements, a per-element output) and its training recipe, and take z-scored inputs, gradient clipping and best-validation checkpointing from its PyTorch code. Only the input changes: per element, the cascaded channel $g_n\mathbf{F}_n$ and the direct channel, $4N_t$ real values. With $N_t > 1$ the optimum is defined only up to a common phase, which the weak direct link barely fixes, so the labels set element 1 to zero; the common phase of a prediction is restored in closed form, $\phi = \angle(\mathbf{c}^H\mathbf{h}_d)$, with $\mathbf{c}$ the cascaded channel of the predicted phases. The network predicts one user's phase, so we use $K = 1$, a maximum-ratio precoder and rate $\log_2(1 + P\|\mathbf{h}\|^2/\sigma^2)$. Its exact element-wise maximizer, coordinate ascent (CD) with closed-form element updates, provides both the labels and the classical baseline.

On 6000 static test scenarios of the training draw, the network reaches a phase RMSE of 61.32° against the labels (24.9% of elements within 5°) and a rate of 12.490 b/s/Hz, between SU alignment (12.431) and the optimum (12.893); it closes 12.8% of that gap. The flattened head of the PyTorch code gives 66.45°, and 83.84° with separate $\mathbf{F}$ and $\mathbf{g}$ as input. As a control, on the published single-antenna task ($N_t = 1$, label $-\angle(g_nF_n)$) the same code reaches 14.99° in our generator, against 2.77° reported in [14] for its own channel generator; absolute errors are therefore not comparable across the two studies, and with $N_t = 1$ the closed form is exact

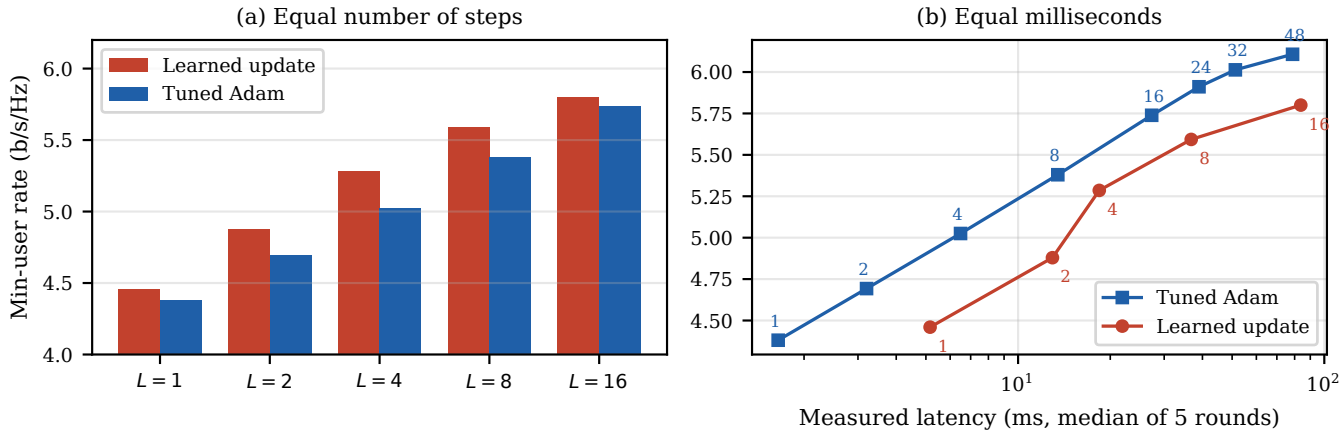


Fig. 5. Learned update rule versus tuned Adam on the static test set. (a) At an equal number of steps, the learned rule is ahead at every budget. (b) On the measured latency axis (numbers: steps), every learned configuration lies below and to the right of the Adam curve, i.e., it is dominated.

TABLE VIII
SINGLE-USER SET (400 SCENARIOS): LATENCY, $t = 0$ RATE AND EFFECTIVE RATE (B/S/HZ) OF THE PORTED CNN–LSTM AND OF CLASSICAL METHODS. BOLD: BEST METHOD AT EACH T_{ov}, SELECTED ON A SEPARATE DRAW. ADAM'S STEP SIZE SELECTED ON VALIDATION SCENARIOS AT EACH L; SHOWN ARE $L = 1$ AND THE BEST L AT $T_{ov} = 0$. RANGES IN THE TEXT ALSO INCLUDE $T_{ov} = 1.71$ MS.

Method	ms	t=0	Effective, T_{ov} (ms) 0	3.43	10
SU alignment	0.04	12.389	12.389	12.369	12.249
CD, 1 sweep	2.48	12.709	**12.698**	**12.656**	**12.491**
CD, 2 sweeps	4.76	12.736	12.701	12.638	12.441
CD, 10 sweeps	23.06	12.742	12.152	12.047	11.984
Adam, L=1	0.59	12.499	12.498	12.472	12.339
Adam, L=8	3.89	12.690	12.664	12.611	12.427
CNN–LSTM	1.12	12.355	12.354	12.326	12.193
CNN–LSTM + CD, 1 sweep	4.26	12.700	12.671	12.614	12.424
CD, converged (offline)	–	12.745			

(10.349 b/s/Hz, network 10.334). With $N_t = 8$, the optimal phase of each element depends on the beam formed by all others, and the network captures little of this coupling.

Table VIII gives the effective rates. At $t = 0$ the network cannot be distinguished from SU alignment on these 400 scenarios (difference -0.034 b/s/Hz, paired 95% CI -0.094 to 0.026); on two larger static draws of 6000 scenarios it is slightly above it ($+0.059$ and $+0.040$, both CIs above zero). It is, however, dominated by a single Adam step, which takes 0.59 ms against the network's 1.12 ms and is better by 0.143 b/s/Hz at $t = 0$ (CI 0.089–0.197), and by 0.090 and 0.104 on the larger draws. At every T_{ov} the network is 0.297–0.344 b/s/Hz below the best method, CD with one sweep, selected on a separate 400-scenario draw (paired CIs exclude zero). As the start of one CD sweep the network adds nothing at $t = 0$ (-0.009, CI -0.021 to 0.004) and, at higher latency (4.26 versus 2.48 ms), ends 0.028–0.066 b/s/Hz below the best. Ten CD sweeps raise the $t = 0$ rate to 12.742, but the effective rate falls to 12.152. In this multi-antenna scenario, one forward pass buys less than one classical step of half the cost; [14] evaluated a static single-antenna channel.

TABLE IX
FRESH VERSUS STALE PRECODER AT $\tau = T_c = 28.2$ MS, PHASE FIXED AT $t = 0$ IN BOTH (BEAM SET, 400 SCENARIOS).

Phase chosen at $t = 0$	Fresh beam (b/s/Hz)	Stale beam (b/s/Hz)	Ratio
SU-average	3.586	0.306	11.7×
SU-best-of	3.924	0.305	12.9×
Adam, $L = 8$	4.358	0.307	14.2×

H. What the Fresh-Beam Assumption Is Worth

All results above assume that the BS precoder is recomputed from the current channel while only the RIS phase ages (Section IV). Table IX measures the alternative in which the precoder ages too, on the separate beam set (Section V), whose absolute rates are not comparable with Table I. Both curves use the same $t = 0$ phase. With a fresh beam the rate decays moderately (Adam with $L = 8$: from 5.342 to 4.358 b/s/Hz at T_c); with a stale beam it collapses within tens of milliseconds, and at T_c it is 11.7–14.2× lower than with a fresh beam, for every phase choice. The assumption therefore matters, and it is demanding: with the phase of Adam ($L = 8$), a precoder only 5 ms old already lowers the rate from 5.228 to 1.736 b/s/Hz. For the effective rates above to apply, the BS has to recompute its precoder within a few milliseconds of every change, which requires effective-channel pilots at that rate; their airtime is not charged in our results. Refreshing it is cheap: single-threaded, computing the effective channel and the regularized ZF precoder takes a median of 0.12 ms, about 13× less than a single Adam step (8.0–15.0× across five rounds). We have not included this time in the effective rates. This is the usual frame-by-frame operation of a multi-antenna BS, but it is an assumption of our results rather than a finding.

VII. CONCLUSION

When users move, an RIS phase configuration is worth what it achieves on the channel that exists when it is delivered. Evaluated this way on a 3GPP indoor-office multi-user MISO scenario with pedestrian users, the methods with the best rate

at $t = 0$ take 0.9–5.9 s and end below a 2.5 ms single-user heuristic, and the number of optimizer steps has an interior optimum: Adam's effective rate peaks at 4–6 steps, and running it to $L = 48$ costs 14.0%. The optimum is governed by the normalized step cost $x = \Delta T f_d$. It moves from 96 steps to 2 as x grows over two and a half decades, and a small-delay model explains the scaling, $L^\star \propto x^{-2/(\gamma+2)}$, consistently with the measured exponent; a 64-element surface has the same optima. Whether convergence hurts therefore depends on how fast the optimizer runs relative to how fast the channel changes. In a controller that re-optimizes periodically, the pilot overhead of each update sets how often to update, while a few warm-started steps per update remain best. A learned update rule that beats a tuned Adam at every equal step count is dominated by Adam at every measured latency, and a published phase-aware CNN–LSTM, ported to a single-user version of the scenario, is dominated by a single Adam step. Latency does not always decide the ranking, but in the regime quantified here it reverses it.

A. Practical Implications

- Choose the number of steps from $x = \Delta T f_d$ and the acquisition delay, not from convergence; (12) gives a first estimate from two fitted constants. In the setup studied here, four to six steps (7–10 ms) are best for a single update, and two to eight warm-started steps per update in periodic operation.
- Choose the update period from the pilot overhead, warm-start each update, and keep each update short.
- Compare methods on measured latency, with the classical baseline's hyper-parameters tuned at each budget, and report dominance rather than equal-step wins.
- Measure latency only on an idle machine, report it relative to $1/f_d$, and check that the verdict holds in every repeated round.

B. Limitations and Future Work

The channel model keeps scatterers fixed, so effective rates beyond about 200 ms are optimistic, and draws scattered-path gains independently per BS antenna, which makes all rates optimistic. Channel knowledge is perfect apart from the acquisition delay and pilot overhead we add; RIS switching time is not modelled; the model is narrowband and far-field, with one geometry, $N \in \{32, 64\}$ and $f_c = 3$ GHz. The normalized step cost carries the results to other speeds and approximately to other carriers, but not path loss. The learned update rule was compared with Adam only on the static set, and the ported CNN–LSTM only for one user; the cost ratio on hardware other than a single CPU thread remains to be measured. Coordinate descent and long-run Adam were run at fixed budgets, and fewer sweeps were evaluated only for a single user. Semidefinite relaxation [2], WMMSE-type algorithms [10] and manifold optimization were not timed. The results also rely on the BS recomputing its precoder from the current channel within a few milliseconds, with effective-channel pilots whose airtime we do not charge. Natural extensions for the multi-user problem are imperfect CSI, with clean-trained and noise-trained models reported separately, discrete phases (1–3 bits) with the latency of discrete optimizers, and wideband, near-field and vehicular operation.

Appendix
Measurement Traps

Three errors produced wrong numbers during this work, and each led to a rule in Section V. (i) A timing of Adam with $L = 8$ taken while a training job was running gave 29.70 ms, against 17.48 ms without it (project log), and put a learned configuration on the latency–rate front; timings are now taken with no other job running, the load average is recorded where the script allows it, and dominance is reported only if it holds in every round. (ii) An interactively computed ratio of fresh to stale beam, "17×", is 11.7–14.2× when measured with a script (Table IX); every number now comes from a script. (iii) A summary that had not been regenerated after a re-measurement listed older timings; tables are generated directly from the result files.

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Use of AI tools. In accordance with IEEE policy, the authors disclose that the AI assistant Claude (Anthropic), used through the Claude Code tool, was used to draft and revise the text of this article, to write the analysis, plotting and table-generation scripts and the additional experiment scripts reported here, to generate the figures and tables from the stored measurement results, and to search for and check references. Earlier AI-generated drafts contained errors, including references that do not exist; these were removed. Apart from values quoted from earlier work or the project log, every number in this article is produced by a script from stored measurement files, every reference was checked against its publisher record, and automated independent audits compared the text with the code and data. [Confirm, and state whether AI was also used for the simulation code in ris_mu/.] The authors reviewed all AI-generated content and take full responsibility for the article.

Data and Code Availability

The scripts that generate every table and figure, and the stored measurement files they read, are available at [public GitHub repository URL].

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